GitHub EMU adoption proposal

Stakeholders and sponsors | hi3g-enterprise | Approximately 189 users | Five organizations

Status: Draft for decision. Based on the described environment, not a tenant inspection. Commercial terms, the Copilot offering and the complete domain list remain unconfirmed.

Executive recommendation

Approve a gated transition to company-managed GitHub identities using the existing Windows on-premises Active Directory (AD) connection to Microsoft Entra ID. Retain the five-organization business structure, automate membership through approved groups, and replace organization-wide Write access with repository-specific team permissions.

Why change

Most users currently use personal GitHub accounts with Entra SSO. SSO was described as configured but not enforced. All organizations grant Write as their base permission; existing teams do not provide repository-level isolation. Account ownership, access scope and offboarding should be brought under corporate governance.

Decision requested now

Approval	Scope
Discovery and pilot	Assign a sponsor, delivery lead, IAM/platform capacity and application owners.
Target policies	Managed identities, mandatory corporate authentication, private repositories and least privilege.
Commercial engagement	Obtain written GitHub confirmation of Enterprise/Copilot entitlement and transition costs.
Production gate	Approve production waves only after pilot evidence, final costs and recovery readiness.

Key conditions

New enterprise account does not automatically mean a second permanent subscription. The target is one replacement production enterprise; the existing agreement's application and temporary overlap must be confirmed with GitHub.

Plan for no unplanned loss of required access or failed builds/deployments, not guaranteed zero interruption. Short, agreed write-freeze windows may be needed.

Planning baseline: 12 weeks. Rebaseline after repository and integration discovery.

1. Proposed solution and access policy

Identity chain: Windows on-premises AD -> existing AD synchronization -> Entra ID (identity provider) -> GitHub EMU through authentication and SCIM provisioning.

Requirement	Target control
No personal-account access	Use only Entra-provisioned managed accounts in the target. Remove source access after approved cutover/retention steps.
Corporate SSO and domains	Mandatory Entra authentication; provisioning eligibility checks for approved corporate identities and email attributes.
Only assigned repositories visible	Private repositories + base permission None + explicit team grants; reconcile alternate access paths.
Group-driven lifecycle	AD-owned group changes sync to Entra; SCIM updates users/groups; linked GitHub teams provide approved entitlements.

Corporate identity eligibility

Approve the exact domain register, including tre.se, 3.dk and the confirmed German domain (the supplied 'oiser.de' spelling requires validation). Do not assume mail and sign-in UPN are identical. Block missing or unapproved attributes through the selected IAM/provisioning eligibility process and monitor later changes. Employees do not self-register their managed GitHub accounts; Entra creates them.

Repository permissions

Internal repositories are readable by regular enterprise members even with base None. Use private repositories to meet the visibility requirement. A lower team role cannot subtract a higher base or direct grant. Ordinary users should see only their approved private repositories; organization owners and other privileged roles can have broader access.

Reuse groups, but verify their scope

Reuse existing groups where they are appropriate project entitlements. Organization-wide groups alone cannot provide project isolation. Link supported Entra security groups with direct membership to flat GitHub teams: nested memberships and Microsoft 365 groups are not supported for EMU team synchronization. GitHub still defines team-to-repository roles.

Keep administrator roles separately approved. For a leaver, removing one team group is not full offboarding: withdraw application eligibility or deactivate the identity, verify SCIM suspension and check remaining access paths. Measure both synchronization stages.

2. Enterprise and Copilot commercial position

What 'new EMU enterprise' means

GitHub's documented adoption path from an enterprise using personal accounts requires migration to a new managed-user enterprise account. It is a replacement environment, not an in-place conversion of existing personal accounts. EMU is an Enterprise Cloud identity option, not a separate EMU per-user add-on.

Cost area	Position requiring confirmation
Enterprise entitlement	Ask GitHub how current contracted seats and terms apply to the replacement account. Do not purchase another permanent subscription by default.
Temporary coexistence	One Enterprise plan entitles one Enterprise Cloud deployment. Obtain overlap duration, charges and duplicate-seat treatment in writing.
Migration delivery	Budget internal staff effort; obtain a quote for any optional GitHub Expert Services.
Platform usage	Assess Actions, storage, runners and other usage. Target testing may add consumption.
Entra controls	Confirm existing entitlements cover intended authentication, provisioning and access-governance controls.

Protect the existing Copilot investment

Managed accounts support enterprise-funded Copilot. Preserve the current offering unless a change is explicitly agreed; EMU alone does not require an upgrade from Copilot Business to Copilot Enterprise. The exact current plan, assigned-seat count and billing model have not been inspected; 189 GitHub users does not imply 189 Copilot seats.

New managed identities require target Copilot access assignments and IDE reauthentication. Pilot Copilot before removing old assignments. Confirm how contracted entitlements, usage budgets, policies and overlap billing will be handled. Do not assume assignments transfer automatically or that personal and managed identities are deduplicated for billing.

Request to the GitHub account manager

We have hi3g-enterprise, five organizations, approximately 189 GitHub users and an existing enterprise-funded Copilot subscription. Please confirm the supported EMU migration path, application of current Enterprise and Copilot entitlements to the replacement enterprise, temporary coexistence duration and charges, duplicate-seat treatment, naming options and any migration-service costs.

Budget status: no vendor quote reviewed; no guaranteed saving or final cost asserted. Approval of discovery/pilot does not authorize an unquoted production purchase.

3. Continuity and implementation strategy

Build in parallel; keep the source authoritative

Initially leave source access, SSO, base permissions, credentials, runners and integrations unchanged. Provision target managed identities and prepare approved permissions separately. Any interim source hardening must be a separate, tested and approved change. Retain users' existing access until their migration wave is accepted.

Impact	Preparation and mitigation
User identity and Git/IDE access	Provide account-switching guidance, target credentials and Copilot testing before cutover.
Actions and deployments	Restore secrets/variables/environments; prepare runners, apps, webhooks, deployment trust and cross-repository dependencies.
URLs and organization names	Confirm naming with GitHub; update remotes, cloud OIDC claims, CODEOWNERS, packages and external integrations.
Data and configuration	Rehearse the chosen importer; assess LFS, packages, Projects, Discussions, rulesets, attribution and audit retention separately.
Public/partner collaboration	Managed users cannot contribute outside the enterprise; approve a separate governed workflow.

Avoid duplicate production execution

Keep target production schedules, webhooks and deployment triggers disabled during rehearsal. Use isolated test environments. Source production automation remains authoritative until the wave is accepted; do not simultaneously deploy from source and target.

Controlled cutover for each organization

1. Confirm target identities, access, integrations and recovery readiness.
2. Notify users, agree the window and let active builds/deployments complete.
3. Pause new source writes and conflicting triggers; perform final migration.
4. Restore required configuration and validate data, access and critical pipelines.
5. Obtain owner approval, then enable target automation and target writes.
6. Restrict the source under the approved recovery/retention policy.

Only one production write/deployment authority at a time. Returning to the source after target writes requires reconciliation, not a blind switchback. Directory/SCIM suspension cannot erase code already cloned; endpoint and data-handling controls remain necessary.

4. Several-week delivery plan

Provisional 12-week baseline; elapsed weeks start after authorization and necessary access. Repository volume, integration complexity and vendor arrangements may extend the schedule.

Week	Activity	Exit evidence
1	Vendor/commercial and architecture confirmation	Supported path, overlap and ownership agreed
2	Inventory users, grants, repositories and automation	Identity/dependency register complete
3	Approve domain, group and repository-access design	Application owners approve intended access
4	Build EMU authentication, SCIM and recovery	Target identity foundation works
5	Prepare small-org pilot and target integrations	Rehearsal ready; source unchanged
6	Rehearse migration, access, lifecycle and pipelines	Pilot acceptance tests pass
7	Cut over pilot organization and provide hypercare	Pilot users and production workflows accepted
8	Resolve issues; finalize costs and wave runbooks	Sponsor approves remaining rollout
9-10	Migrate remaining four organizations in waves	Each organization owner accepts cutover
11	Stabilize and reconcile access/credentials	No unexplained residual dependency
12	Retention, retirement and operational handover	Business/security/commercial closure

Pilot selection

Choose a low-risk organization with approximately 10-15 representative users where practical, multiple entitlement teams and at least one realistic build/deployment path. Include different corporate domains, a multi-organization user and applicable Copilot usage.

Schedule discipline

Rehearsal determines cutover duration; do not promise a fixed outage window beforehand. Pause later waves if critical access or deployment defects remain unresolved. Retirement is conditional on retention, recovery and commercial approval, not solely week 12.

5. Acceptance, ownership and sponsor gates

Acceptance check	Required result
Identity	Personal-account target access denied; approved managed SSO works; unapproved-domain provisioning blocked.
Repository isolation	Assigned private repo accessible; unassigned repo absent from lists/search and denied by direct URL, API and clone.
Lifecycle	Group addition/removal gives intended access; full offboarding reaches SCIM suspension; propagation times measured.
Operations and Copilot	Critical build/deployment, runners and integrations pass; pilot IDE/Copilot access works before old assignment removal.
Cutover and recovery	Data/configuration reconciled; one production authority; recovery procedure tested and owner accepted.

Accountability

Sponsor: funding, priorities and gates. Delivery/platform lead: migration and configuration. IAM team: AD/Entra, eligibility and SCIM. Security: controls, exceptions and monitoring. Application owners: entitlement and deployment acceptance. Procurement/vendor manager: Enterprise/Copilot terms. Service desk: account-transition support.

Success measures and gates

Target: all ordinary human production access through managed identities; private restricted repositories; base None in five organizations; no routine manual invitations; owned groups and no unexplained direct grants. Preserve a documented setup/recovery exception and separately governed non-human credentials. Agree onboarding/revocation objectives only after pilot measurement; review privileged access monthly and entitlements quarterly.

Gate 1: approve discovery and pilot. Gate 2: approve production after costs, pilot evidence and recovery readiness. Gate 3: approve each organization cutover. Gate 4: approve source retirement and operational closure.

Official references

Enterprise type and migration requirement: https://docs.github.com/en/enterprise-cloud@latest/admin/concepts/enterprise-fundamentals/choose-an-enterprise-type

Deployment entitlement: https://docs.github.com/en/migrations/overview/migrations-within-github

Copilot licensing: https://docs.github.com/en/billing/concepts/product-billing/github-copilot-licenses

Groups and linked teams: https://docs.github.com/en/enterprise-cloud@latest/admin/managing-iam/provisioning-user-accounts-with-scim/managing-team-memberships-with-identity-provider-groups

Migration coverage: https://docs.github.com/en/migrations/using-github-enterprise-importer/migrating-between-github-products/about-migrations-between-github-products